

Monolithic Systems

Definition

A **monolithic system** is an operating-system structure in which the entire operating system runs as one large program in **kernel mode**. All operating-system procedures are linked together into a single executable binary.

1. Basic Idea

Main Concept

In the monolithic approach:

- the whole operating system runs in kernel mode,
- it is built as one large executable program,
- all procedures are linked together,
- one procedure may call another directly.

Efficiency

This design is very efficient because any operating-system procedure can call any other useful procedure directly.

2. Main Characteristics

Procedure Organization

The operating system is written as a collection of procedures. These procedures are compiled and linked together to form one large object program.

Features

A monolithic system usually has:

- one large kernel,
- many procedures inside it,
- unrestricted procedure calls,
- high execution efficiency.

3. Advantages of Monolithic Systems

Advantages

Main advantages are:

- fast communication between procedures,
- simple direct procedure calls,
- efficient execution,
- low overhead.

4. Disadvantages of Monolithic Systems

Disadvantages

A monolithic system may also have important drawbacks:

- the system may become very large and hard to understand,
- thousands of procedures may interact without restriction,
- debugging becomes difficult,
- a crash in one procedure may crash the entire operating system.

5. Building the Monolithic Kernel

Construction

To build the operating system in this model:

1. all individual procedures or files are compiled,
2. all compiled parts are linked together,
3. the linker produces one executable kernel program.

Information Hiding

In a pure monolithic system, there is very little information hiding. Almost every procedure can see and call every other procedure.

Comparison

This is different from modular systems, where much of the internal information is hidden and only specific entry points are visible.

6. Internal Structure in a Monolithic System

Basic Structure

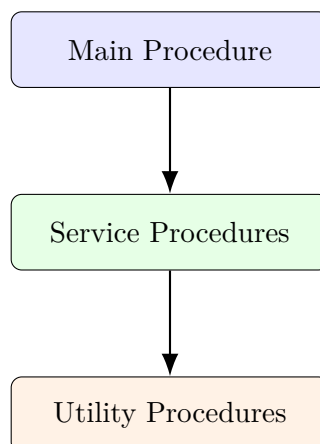
Even in a monolithic system, some internal organization is possible. A simple structure contains:

1. a **main program**,
2. a set of **service procedures**,
3. a set of **utility procedures**.

Roles

- The **main program** receives the request and invokes the correct service procedure.
- The **service procedures** perform the actual system calls.
- The **utility procedures** help the service procedures by doing common supporting work.

6.1 Simple Structure Diagram



7. System Call Handling

System Call Flow

In a monolithic system, a system call is requested by placing parameters in a known place, such as the stack, and then executing a **trap instruction**.

Steps

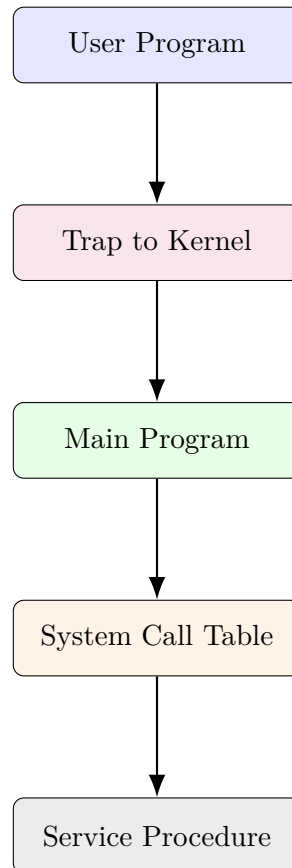
The basic sequence is:

1. user program places parameters,
2. trap instruction switches from user mode to kernel mode,
3. operating system checks which system call is requested,
4. system indexes into a table,
5. the correct service procedure is called.

Table Lookup

The operating system often uses a table where entry k contains a pointer to the procedure that handles system call k .

7.1 System Call Dispatch Diagram



8. Loadable Extensions

Extensions

In addition to the main kernel loaded at boot time, many operating systems support **loadable extensions**.

Examples

These extensions may include:

- I/O device drivers,
- file systems,
- additional kernel support components.

Loaded on Demand

These components are often loaded only when needed, rather than being permanently included in the core kernel image.

9. Shared Libraries and DLLs

Names in Different Systems

Different operating systems use different names for such loadable extensions:

- In UNIX, they are often called **shared libraries**.
- In Windows, they are called **DLLs** (Dynamic-Link Libraries).

Windows DLL

Windows DLL files usually have the extension:

.dll

10. Summary Points

Quick Review

A monolithic system:

- runs entirely in kernel mode,
- is built as one large executable,
- allows direct calls among procedures,
- is efficient but less safe and harder to manage,
- may still have internal structure through main, service, and utility procedures.